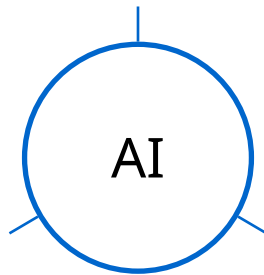


Medical Billing Automation Agent

n8n Workflow Architecture & Implementation Guide



Intelligent Claim Processing

Prepared for: Healthcare Providers & Clinics

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December 9, 2025

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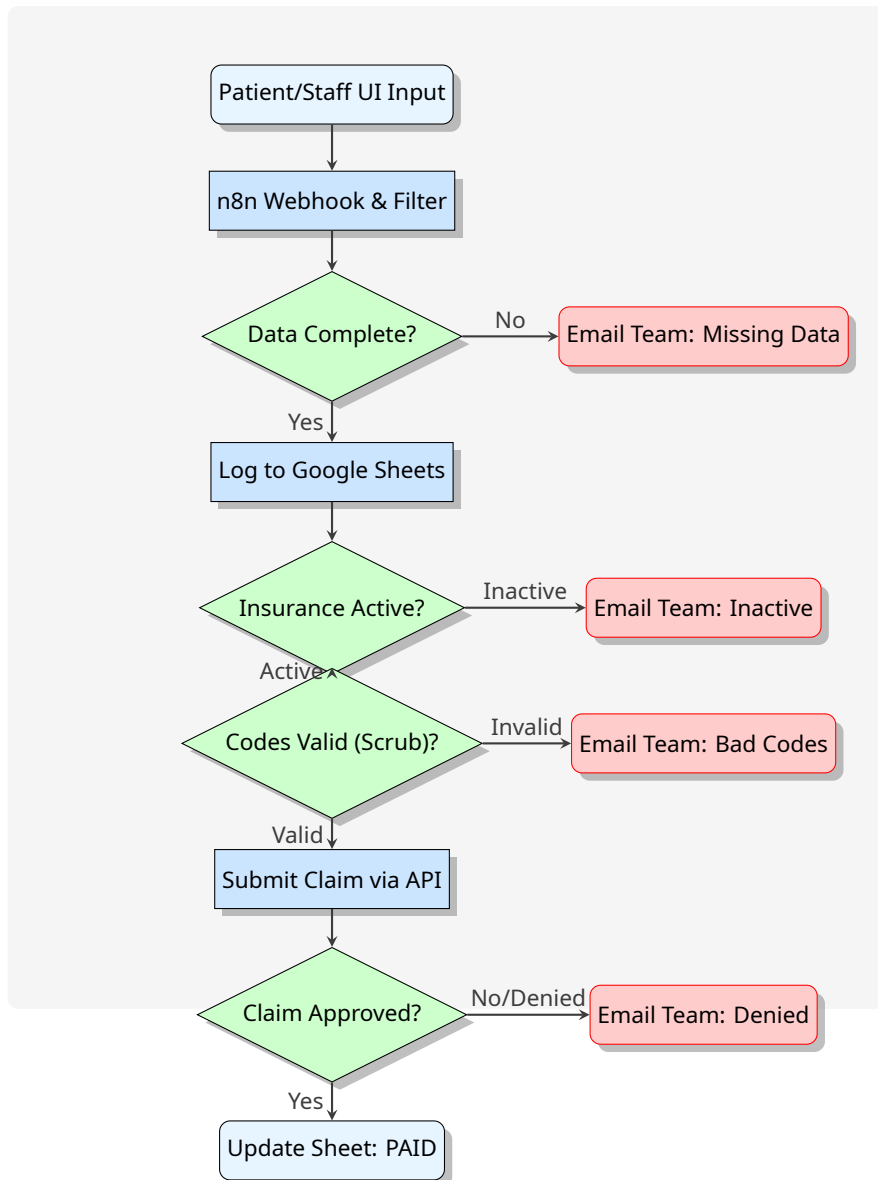
1 Executive Summary

This document outlines the architecture of our ****AI-Powered Medical Billing Automation Agent****. Designed using n8n, this system streamlines the complex process of patient insurance claims, reducing human error, speeding up reimbursement cycles, and ensuring compliance through rigorous data validation and "claim scrubbing."

The system integrates a user-friendly frontend for data entry (or EHR integration) with a robust backend that handles validation, verification, and submission to clearinghouses automatically.

2 Automation Workflow Diagram

The following diagram illustrates the logic flow of the agent, from patient intake to final claim approval.



3 Technical Process Breakdown

3.1 1. Data Ingestion

The process begins when patient data is entered into the User Interface. The system captures the following critical data points:

- Patient Name & Date of Birth (DOB)
- Insurance Company Name & Member/Policy ID
- Date of Service & Provider Name
- ICD-10 (Diagnosis) Codes
- CPT (Procedure) Codes
- Total Charge Amount

This data is transmitted securely via **Webhook** to our n8n environment.

3.2 2. Filtering & Logic Engine

Once received, the automation performs a filtering operation to strip unnecessary meta-data, isolating only the clinical and financial data required for processing. A **Code Node** then restructures the JSON data into a standardized format accepted by insurance APIs.

3.3 3. Automated Validation (The "If" Node)

Before any external API calls, the system performs an internal audit:

- **Check:** Are all required fields present?
- **Action:** If data is missing, the workflow halts, and an alert email is sent to the administrative team for correction.

3.4 4. Verification & Scrubbing

1. **Eligibility Check:** The system queries the insurance provider's API. If the policy is inactive, the claim is flagged immediately.
2. **Claim Scrubbing:** The system analyzes the ICD and CPT codes. It checks for correct structure (digits/format) and compatibility. Incorrect codes trigger an immediate alert, preventing a formal denial later.

3.5 5. Submission & Status Tracking

Valid claims are submitted via the Clearinghouse API. The system monitors the response:

- **Approved:** The Google Sheet is updated with the "Approved" status.
- **Denied:** The status is logged as "Denied" with the error reason, and the team is notified via email.

EHR Integration Capability

Note to Clients: While we provide a custom User Interface for data entry, this automation is fully capable of connecting directly to your hospital's **Electronic Health Record (EHR)** system. We can fetch patient data automatically via API (HL7/FHIR), eliminating manual entry entirely.

4 Benefits for Hospitals & Clinics

- **Reduced Rejection Rate:** By "scrubbing" claims for errors before submission, we drastically lower the chance of denial.
- **Labor Savings:** Staff no longer need to manually verify eligibility or type data into clearinghouse portals.
- **Faster Revenue Cycle:** Instant submission and immediate error detection mean faster payments from insurance companies.

5 Cost Estimation & Service Model

Below is the financial breakdown for the implementation and ongoing operation of this automation.

5.1 Detailed Cost Breakdown

Service Component	Description	Estimated Cost
Initial Setup & Development	Workflow design, n8n setup, and validation logic config.	\$1,500 - \$3,000 (One-time)
API Integration (Monthly)	Ongoing connection maintenance for Insurance APIs/EHR.	\$100 - \$200 (Per Month)
System Maintenance	Server monitoring, bug fixes, and minor updates.	\$200 - \$500 (Per Month)
Usage Fees (Pass-through)	Clearinghouse API calls and n8n hosting costs.	Variable (≈\$50/mo)

Table 1: Investment Components

5.2 Investment Summary

To assist in your budgeting, we have aggregated the costs into "Building" (One-time) and "Running" (Monthly) categories.

Total Investment Required

Total Building Cost (One-Time):

\$1,500 – \$3,000

Includes full development, testing, and deployment.

Total Monthly Cost (Recurring):

\$250 – \$350

Includes API integrations, system maintenance, and estimated usage fees.

Disclaimer: Please note that these figures are estimates. The exact cost depends on your specific system requirements, transaction volume, and existing infrastructure. This is a temporary projection for planning purposes.